

2.4 Continuity

Definition 1: A function f is continuous at $x=a$ if

(i) $f(a)$ exists

(ii) $\lim_{x \rightarrow a} f(x)$ exists

(iii) $f(a) = \lim_{x \rightarrow a} f(x)$

(i) + (ii) + (iii)
(all three conditions)

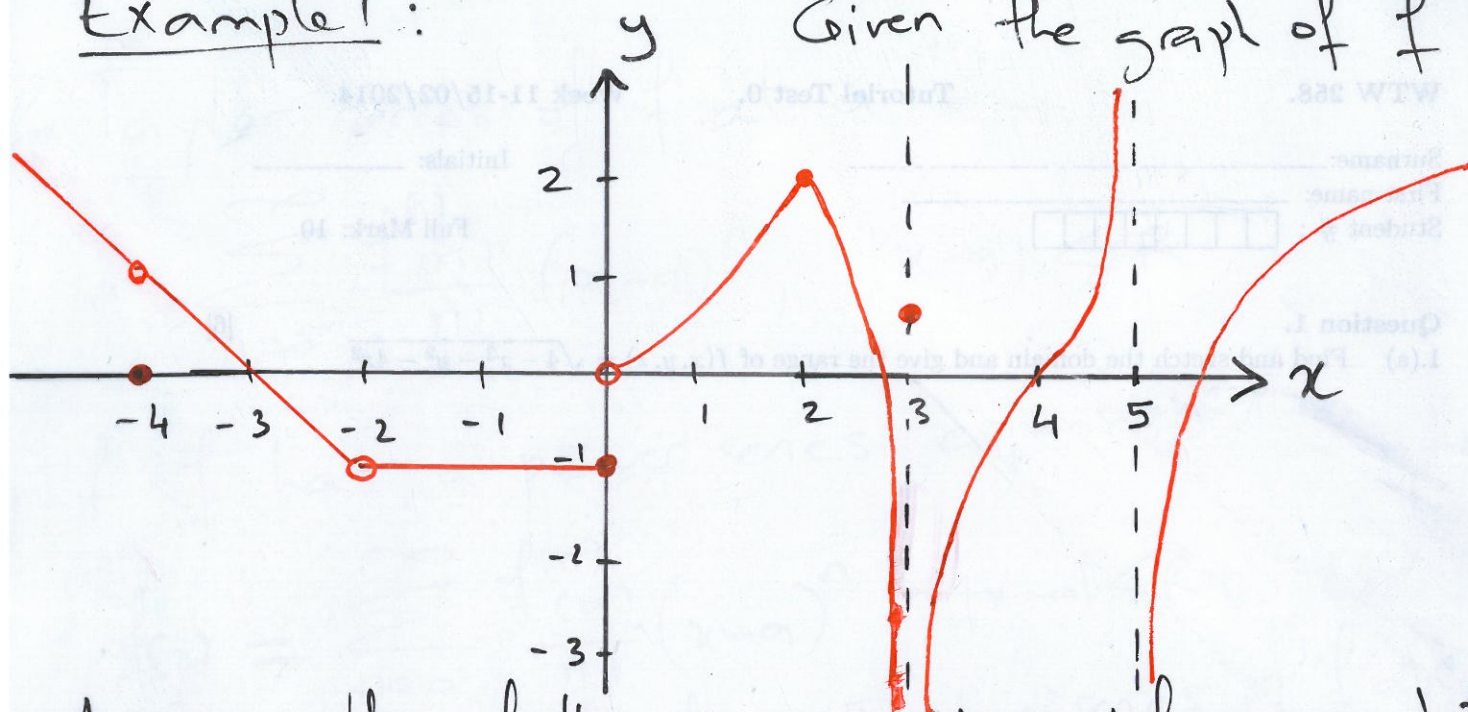
What does it mean graphically?

The graph of f is one piece "near" $x=a$
(use a small open interval around and this interval is getting smaller and smaller)

Definition 2: If f is not continuous at $x=a$ then we say that f is discontinuous at a .

Example 1:

Given the graph of f



Answer the following and justify your statement

(1) Is f continuous at $x = -2$?

(2) _____ $x = 0$?

(3) _____ $x = 2$?

(4) _____ $x = 3$?

(5) _____ $x = 5$?

(6) _____ $x = -4$?

There is three kind of discontinuity at $x=a$

- removable discontinuity (Eg: (6) & (11))

- jump discontinuity (Eg: (2))

- infinite discontinuity (Eg: (4) & (5))

Example 1:

(1) $f(-2)$ does not exist $\Rightarrow f$ not continuous at $x = -2$

(2) $\lim_{x \rightarrow 0^-} f(x) = 1 \neq 0 = \lim_{x \rightarrow 0^+} f(x)$ ($\lim_{x \rightarrow 0} f(x)$ does not exist)

$\Rightarrow f$ discontinuous at $x = 0$

(3) $f(2) = 2$, $\lim_{x \rightarrow 2} f(x) = 2$

$\Rightarrow f$ is continuous at $x = 2$

(4) $\lim_{x \rightarrow 3} f(x) = -\infty$ (does not exist)

$\Rightarrow f$ discontinuous at $x = 3$

(5) $f(5)$ does not exist or $\lim_{x \rightarrow 5} f(x)$ does not exist

$\Rightarrow f$ not continuous at $x = 5$

(6) $f(-4) = 0$, $\lim_{x \rightarrow -4} f(x) = 1 \neq f(-4)$

$\Rightarrow f$ discontinuous at $x = -4$

Eg 1: Characterize the discontinuities

At $x = -2$ and $x = -4$: removable discontinuities

At $x = 0$: jump discontinuity

At $x = 3$ and $x = 5$: infinite discontinuities